

Surrogate assisted diversity estimation in neural ensemble search

Alexandr Udeneev¹, Petr Babkin¹, and Oleg Bakhteev²

¹ Moscow Center for Advanced Studies, 20 Kulakova St., Moscow, Russia

² Applied AI Institute, Moscow, Russia

{udeneev.av, p.k.babkin, bakhteev.o}@gmail.com

Abstract. Ensembles are a standard way to improve the performance and robustness of deep neural networks, but their effectiveness crucially depends on both the quality and the diversity of individual models. Most neural architecture search (NAS) methods are computationally expensive. Extending them to neural ensemble search (NES), which requires joint optimization of individual architectures and their ensemble composition, leads to an exponential growth of the search space and makes the problem computationally intractable. To address this, we introduce a dual-objective surrogate-guided ensemble search: candidate architectures are represented as directed acyclic graphs, and two surrogate models are trained independently to estimate predictive accuracy and diversity potential. Their combined estimates guide an NES framework that efficiently identifies architectures that are both individually strong and collectively diverse. Our final ensemble achieves competitive accuracy with state-of-the-art methods on FashionMNIST, CIFAR-10, and CIFAR-100, surpassing standard baselines such as Deep Ensembles and Random Ensemble Search.

Keywords: neural ensemble search · ensemble diversity · surrogate function · triplet loss.

1 Introduction

Neural network ensembles consistently demonstrate improved predictive accuracy, robustness, and uncertainty estimation compared to single models, particularly in classification and regression tasks [1, 2]. This observation motivates the study of *Neural Ensemble Search* (NES), which aims to construct ensembles of neural networks in a principled and computationally efficient manner [3, 4]. In practice, NES relies on *Neural Architecture Search* (NAS) techniques to identify suitable candidate architectures, a task that remains challenging due to the vastness of the search space and the high cost of training individual models [5].

A widely used baseline for ensemble construction is Deep Ensembles (Deep-Ens) [6], which combine multiple independently trained instances of the same architecture. While simple and effective, this approach offers limited control over ensemble diversity, often resulting in highly correlated models. More recent NES

methods adapt NAS techniques to ensemble construction [4, 7], but diversity in these approaches typically emerges implicitly through optimization dynamics or heuristic similarity measures, rather than being explicitly modeled and optimized alongside accuracy.

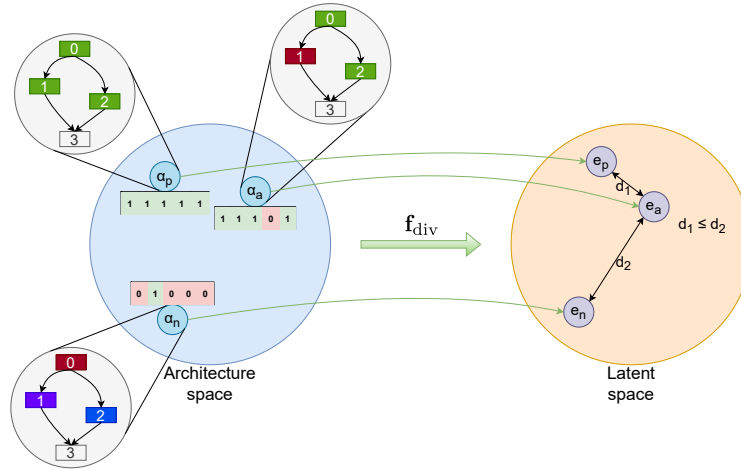


Fig. 1: Illustration of the surrogate diversity function. Neural architectures are embedded into a latent space such that architectures with similar predictive behavior are mapped close together, while dissimilar ones are separated by larger distances. This enables efficient diversity estimation without training candidate models.

To address these limitations, we draw inspiration from surrogate-assisted NAS methods, which employ lightweight predictive models to estimate architecture performance without full training [8, 9]. While surrogates are commonly used to predict accuracy, recent work suggests that they can also capture structural and functional relationships between architectures [10]. This observation motivates their use for explicit diversity modeling in NES.

We propose a dual-objective surrogate-guided framework for Neural Ensemble Search. The framework separates the search into two complementary predictive objectives: accuracy estimation for individual architectures and diversity modeling via latent architecture embeddings. Candidate architectures are encoded as graphs and mapped into a continuous latent space using surrogate models, where geometric distances correspond to predictive dissimilarity. As illustrated in Fig. 1,

this representation enables explicit, differentiable, and training-free evaluation of ensemble diversity. We implement the surrogate models using Graph Attention Networks (GATs) [11] and structure the latent space using the triplet loss [12] for metric learning.

Our main contributions are:

1. We introduce a novel surrogate-based methodology for training diversity predictors that map discrete neural architectures to a structured latent space.
2. We present the surrogate-assisted framework for Neural Ensemble Search that jointly optimizes predictive performance and architectural diversity via dual-objective guidance. To the best of our knowledge, this is the first method to address this problem using surrogate models.
3. We provide a proof-of-concept validation of our framework on FashionMNIST [13], CIFAR-10 [14], and CIFAR-100 [14], demonstrating its capability to balance predictive performance and architectural diversity effectively.
4. We provide datasets of approximately 3,000 trained models per dataset (including both weights and architectures) for FashionMNIST, CIFAR-10, and CIFAR-100³. These datasets enable evaluation of ensemble methods without the need to train large numbers of models from scratch. Training details are provided in Section 4.1.

2 Problem statement

Neural Architecture Search methods have proven effective at discovering high-performing individual models. However, many predictive tasks benefit from ensemble strategies that aggregate multiple models rather than relying on a single architecture. The challenge lies in the fact that an optimal ensemble requires not merely a collection of strong models, but a set of architectures that actively complement one another. This motivates our formulation of Neural Ensemble Search as a unified optimization problem, where architectural selection and ensemble composition are jointly optimized. We begin by reviewing the standard NAS framework, then extend it to the ensemble setting.

2.1 Neural Architecture Search

We define the core components of the search space as:

$$\mathcal{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_N\} \subset \mathbb{R}^d, \quad \mathcal{O} = \{\mathbf{o}_1, \dots, \mathbf{o}_K\}, \mathbf{o}_i : \mathbb{R}^d \rightarrow \mathbb{R}^d,$$

where $\mathcal{V}$ is the ordered set of node feature-vectors (implying a topological order), and $\mathcal{O}$ is a finite set of elementwise operations (e.g., convolutions, poolings).

Using these components, edges are defined as annotated directed relations, represented by tuples of the form $(\mathbf{v}_i \rightarrow \mathbf{v}_j, \mathbf{o})$. We then define the space of

³ The dataset will be published in the camera-ready version of the paper

feasible architectures $\mathcal{A}$ as the set of all labeled directed acyclic graphs, where edges strictly follow the topological order to ensure acyclicity:

$$\mathcal{A} = \{(\mathcal{V}, E) \mid E \subseteq \{(v_i \rightarrow v_j, \mathbf{o}) \mid 1 \leq i < j \leq N, \mathbf{o} \in \mathcal{O}\}\}.$$

Here, the condition $i < j$ restricts edges to forward connections only, intrinsically guaranteeing that every architecture in $\mathcal{A}$ is a Directed Acyclic Graph (DAG).

Consequently, a specific architecture $\alpha \in \mathcal{A}$ can be represented as:

$$\alpha = (\mathcal{V}, \{(v_i \rightarrow v_j, \mathbf{o}_{i,j}) \mid (v_i \rightarrow v_j) \in \mathcal{E}_\alpha\}),$$

where $\mathcal{E}_\alpha$ is the set of active edges satisfying $1 \leq i < j \leq N$, and each edge is uniquely labeled by an operation $\mathbf{o}_{i,j} \in \mathcal{O}$.

Denote $\mathcal{L}_{train}$ and $\mathcal{L}_{val}$ as the training and validation losses, respectively. The NAS problem can then be formulated as the search for an optimal architecture α^* that minimizes $\mathcal{L}_{val}(\alpha^*, \omega^*)$, under the constraint that the weights are obtained by minimizing the training loss:

$$\omega^* = \arg \min_{\omega \in \mathcal{W}} \mathcal{L}_{train}(\alpha^*, \omega).$$

This can be expressed as the following optimization problem:

$$\begin{aligned} \min_{\alpha \in \mathcal{A}} \quad & \mathcal{L}_{val}(\alpha, \omega^*(\alpha)), \\ \text{s.t.} \quad & \omega^*(\alpha) = \arg \min_{\omega \in \mathcal{W}} \mathcal{L}_{train}(\alpha, \omega). \end{aligned} \tag{1}$$

The primary challenge in this optimization lies in the immense search space of possible architectures (e.g., in DARTS [15], it is approximately 10^{25}).

2.2 Neural Ensemble Search

The primary objective of NES is to find an optimal ensemble of neural networks with architectures constrained to the same architectural search space as that explored by NAS methods.

As before, we denote $\alpha \in \mathcal{A}$ as a network architecture and $\omega(\alpha)$ as its corresponding parameters. The output of this network on an input x is denoted by $f_\alpha(x, \omega(\alpha))$. Let $S \subset \mathcal{A}$ denote a subset of architectures. Then, the NES problem can be formally described as follows:

$$\begin{aligned} \min_S \quad & \mathcal{L}_{val} \left(\frac{1}{|S|} \sum_{\alpha \in S} f_\alpha(x, \omega^*(\alpha)) \right), \\ \text{s.t.} \quad & \forall \alpha \in S: \omega^*(\alpha) = \arg \min_{\omega(\alpha)} \mathcal{L}_{train}(f_\alpha(x, \omega(\alpha))). \end{aligned} \tag{2}$$

Thus, in addition to searching over a vast number of architectures, we now also need to find the optimal ensemble composition.

3 Surrogate-based ensemble search

In this work, we adapt the DARTS architecture space representation into a graph-structured form that can be processed by Graph Attention Networks (GATs) (see Section 3.1). In Section 3.2, we present the architecture of the surrogate functions and describe their operating principles. Section 3.3 then details the ensemble construction method based on these surrogate models, explaining how surrogate predictions are used to promote architectural diversity within the ensemble.

Together, these components enable the construction of ensembles that jointly consider predictive accuracy and architectural diversity.

3.1 Architecture Search Space

Following the conventions of [15], we instantiate the search space $\mathcal{A}$ using a cell-based approach. Each architecture $\alpha \in \mathcal{A}$ is composed of two functional units: a *normal cell* α_{norm} and a *reduction cell* α_{red} , such that $\alpha = (\alpha_{\text{norm}}, \alpha_{\text{red}})$. Both units are DAGs adhering to the formalisms established in Section 2.1.

Unlike the original DARTS, which relies on continuous relaxation, our method explores $\mathcal{A}$ via discrete random sampling. We consider cells composed of $n = 5$ nodes and $m = 10$ edges, with each node having exactly two incoming and two outgoing edges. Given an operation set size $|\mathcal{O}| = 7$, the cardinality of the cell-level search space is:

$$|\mathcal{A}_{\text{cell}}| = \prod_{k=2}^{n-1} \binom{k}{2} \cdot |\mathcal{O}|^m \approx 10^9. \quad (3)$$

Consequently, the full architecture space $|\mathcal{A}| = |\mathcal{A}_{\text{cell}}|^2 \approx 10^{18}$ remains computationally intractable for exhaustive search, necessitating surrogate-guided exploration.

To train the dual-surrogate models, we construct a dataset $\mathcal{D}_{\text{train}}$ by evaluating N sampled architectures:

$$\mathcal{D}_{\text{train}} = \{(\alpha_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N, \quad (4)$$

where $\mathbf{y}_i = f_{\alpha_i}(\mathcal{X}_{\text{val}}, \omega^*(\alpha_i))$ is the vector of model predictions on a fixed validation subset $\mathcal{X}_{\text{val}}$, and acc_i is the corresponding validation accuracy. These observations serve as the ground truth for our surrogate functions.

3.2 Surrogate Function

To facilitate the construction of ensembles (Section 3.3), we predict both the performance and diversity of candidate architectures. Due to the enormous size of the architecture space, obtaining these characteristics via full training is infeasible. To address this, we employ surrogate models: given an architectural representation, they predict key properties of neural networks. We consider two surrogates:

$$f_{\text{acc}}^{\theta} : \mathcal{A} \rightarrow \mathbb{R}, \quad \mathbf{f}_{\text{div}}^{\theta} : \mathcal{A} \rightarrow \mathbb{R}^d.$$

Each architecture is represented as a directed acyclic graph (DAG), where nodes correspond to latent representations and edges to operations, similar to NAS-Bench-201 [16]. For surrogate modeling, we adopt an alternative graph representation, inspired by NAS-Bench-101 [17]: nodes represent operations and edges their corresponding latent embeddings. This conversion is performed as follows:

1. Each edge of the original graph is transformed into a node in the new graph, labeled by the operation present on that edge.
2. An oriented edge is drawn from operation $\mathbf{o}_i$ to $\mathbf{o}_j$ if in the original graph $\mathbf{o}_i$ acts from node $\mathbf{x}$ to $\mathbf{y}$ and $\mathbf{o}_j$ acts from $\mathbf{y}$ to $\mathbf{z}$.

Each operation is represented using a one-hot encoding. The surrogates themselves are Graph Attention Networks (GATs) with N sequential layers, residual connections, GraphNorm [18], a global pooling aggregation, and lightweight output heads.

The accuracy surrogate f_{acc} is trained via supervised regression to predict the performance of a given architecture. Given the training dataset $\mathcal{D}_{\text{train}} = \{(\boldsymbol{\alpha}_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N$, we optimize the parameters $\boldsymbol{\theta}_{\text{acc}}$ by minimizing the mean squared error (MSE) relative to the ground-truth validation accuracy:

$$\boldsymbol{\theta}_{\text{acc}}^* = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^N (f_{\text{acc}}^{\boldsymbol{\theta}}(\boldsymbol{\alpha}_i) - \text{acc}_i)^2, \quad (5)$$

In contrast, the diversity surrogate $\mathbf{f}_{\text{div}}^{\boldsymbol{\theta}}$ cannot be trained directly with supervised learning, since ground-truth latent embeddings are unavailable. Instead, we optimize it using a triplet loss on architecture embeddings.

Let us consider a set of I trained models $\{M_1, \dots, M_I\}$ and a fixed validation dataset of K examples. Denote the predictions of model M_i on the validation dataset by the vector

$$\mathbf{y}^{(i)} = (y_1^{(i)}, \dots, y_K^{(i)}).$$

We construct a diversity matrix $\mathbf{C} \in [0, 1]^{I \times I}$, where each entry

$$c_{ij} = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(y_k^{(i)} = y_k^{(j)})$$

measures the fraction of matching predictions between models M_i and M_j .

To facilitate triplet-loss training, the diversity matrix $\mathbf{C}$ is discretized into $\mathbf{D} \in \{-1, 0, 1\}^{I \times I}$ according to upper and lower quantile thresholds q_p and q_n :

$$\mathbf{D}_{ij} = \begin{cases} 1, & \text{if } c_{ij} > q_p, \\ -1, & \text{if } c_{ij} < q_n, \\ 0, & \text{otherwise.} \end{cases}$$

The training procedure for the diversity surrogate $\mathbf{f}_{\text{div}}$ is detailed in Fig. 2. For each training epoch, we sample a batch of triplets $(\boldsymbol{\alpha}_a, \boldsymbol{\alpha}_p, \boldsymbol{\alpha}_n)$ where the

Input: $\mathcal{D}_{train}$: training dataset of architectures $\{(\alpha_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N$;
D: diversity matrix where $\mathbf{D}_{jk} \in \{1, -1\}$ (positive/negative);
 m : triplet loss margin;
 η : learning rate;
 B : batch size.
Output: Optimized parameters θ_{div} for surrogate model $\mathbf{f}_{div}$

```

1 Initialize parameters  $\theta_{div}$ 
2 for epoch  $e \leftarrow 1$  to  $n$  do
3   for step  $t \leftarrow 1$  to  $N/B$  do
4     Sample a minibatch of anchor indices  $\mathcal{J} \subset \{1, \dots, N\}$ , where  $|\mathcal{J}| = B$ 
5     for each  $j \in \mathcal{J}$  do
6       Identify candidate sets:  $\mathcal{P}_j = \{k \mid \mathbf{D}_{jk} = 1\}$  and
7          $\mathcal{N}_j = \{k \mid \mathbf{D}_{jk} = -1\}$ 
8       Sample positive  $k_p \sim \text{Uniform}(\mathcal{P}_j)$  and negative  $k_n \sim \text{Uniform}(\mathcal{N}_j)$ 
9       Compute embeddings:
10         $\mathbf{e}_a, \mathbf{e}_p, \mathbf{e}_n \leftarrow \mathbf{f}_{div}^{\theta_{div}}(\alpha_j), \mathbf{f}_{div}^{\theta_{div}}(\alpha_{k_p}), \mathbf{f}_{div}^{\theta_{div}}(\alpha_{k_n})$ 
11        Compute triplet loss for instance  $j$ :
12         $\ell_j \leftarrow \max(0, \|\mathbf{e}_a - \mathbf{e}_p\|_2^2 - \|\mathbf{e}_a - \mathbf{e}_n\|_2^2 + m)$ 
13      end
14    Compute batch loss:  $\mathcal{L} = \frac{1}{B} \sum_{j \in \mathcal{J}} \ell_j$ 
15    Update parameters:  $\theta_{div} \leftarrow \theta_{div} - \eta \nabla_{\theta_{div}} \mathcal{L}$ 
16  end
17 return  $\theta_{div}$ 

```

Fig. 2: Training procedure for the diversity surrogate function.

relationships are defined by the discretized diversity matrix $\mathbf{D}$. Specifically, for a given anchor architecture α_a , the positive α_p and negative α_n examples are sampled uniformly from the precomputed sets:

$$\mathcal{P}_a = \{k \mid \mathbf{D}_{ak} = 1\}, \quad \mathcal{N}_a = \{k \mid \mathbf{D}_{ak} = -1\}. \quad (6)$$

The architectures are subsequently mapped into a d -dimensional latent space via the surrogate function $\mathbf{f}_{div}$ with parameters θ_{div} :

$$\mathbf{e}_a = \mathbf{f}_{div}^{\theta_{div}}(\alpha_a), \quad \mathbf{e}_p = \mathbf{f}_{div}^{\theta_{div}}(\alpha_p), \quad \mathbf{e}_n = \mathbf{f}_{div}^{\theta_{div}}(\alpha_n). \quad (7)$$

The parameters θ_{div} are optimized by minimizing the triplet loss objective:

$$\mathcal{L}_{div} = \sum_{(a,p,n)} \max\left(\|\mathbf{e}_a - \mathbf{e}_p\|_2^2 - \|\mathbf{e}_a - \mathbf{e}_n\|_2^2 + m, 0\right), \quad (8)$$

where $m > 0$ is a margin hyperparameter that enforces a minimum separation between dissimilar architectures in the embedding space. This optimization scheme encourages the surrogate to learn a manifold where Euclidean distances reflect architectural diversity.

The ground-truth matrix $\mathbf{D}$ used to guide the triplet sampling process is constructed by thresholding pairwise diversity metrics calculated on model outputs. In this work, we primarily utilize the fraction of identical predictions as the diversity measure, although the proposed framework is inherently compatible with other metrics, such as the Jensen-Shannon divergence, Hellinger distance, or the correlation of prediction vectors $\mathbf{y}_i$. These metrics yield an $I \times I$ diversity matrix which is subsequently discretized to facilitate stable triplet sampling and the formation of a structured latent space.

3.3 Ensemble Construction

Once the surrogate models $f_{\text{acc}}^{\theta_{\text{acc}}}$ and $\mathbf{f}_{\text{div}}^{\theta_{\text{div}}}$ are trained, they enable an efficient search for the optimal ensemble without requiring the prohibitive computational cost of training intermediate candidate architectures. The proposed ensemble construction (Fig 3) proceeds in two distinct phases: candidate pool filtering and diversity-driven greedy selection.

```

Input:  $K$ : target ensemble size;
 $N$ : number of initial candidates;
 $\lambda$ : accuracy threshold;
 $f_{\text{acc}}, \mathbf{f}_{\text{div}}$ : trained surrogate models.
Output:  $\mathcal{A}_{\text{best}}$ : selected ensemble of  $K$  architectures.
// Phase 1: Candidate Pool Construction
1  $\mathcal{A}_{\text{cand}} \leftarrow \text{RandomSample}(N)$ 
2  $\mathcal{P} \leftarrow \{(\alpha, a, \mathbf{e}) \mid \alpha \in \mathcal{A}_{\text{cand}}, a = f_{\text{acc}}(\alpha), \mathbf{e} = \mathbf{f}_{\text{div}}(\alpha), a \geq \lambda\}$ 
// Phase 2: Diversity-driven Greedy Selection
3 Select triplet with highest predicted accuracy:  $(\alpha^*, a^*, \mathbf{e}^*) = \arg \max_{(\alpha, a, \mathbf{e}) \in \mathcal{P}} a$ 
4  $\mathcal{A}_{\text{best}} \leftarrow \{\alpha^*\}; \quad \mathcal{E}_{\text{sel}} \leftarrow \{\mathbf{e}^*\}; \quad \mathcal{P} \leftarrow \mathcal{P} \setminus \{(\alpha^*, a^*, \mathbf{e}^*)\}$ 
5 while  $|\mathcal{A}_{\text{best}}| < K$  and  $\mathcal{P} \neq \emptyset$  do
6   For each candidate  $(\alpha_i, a_i, \mathbf{e}_i) \in \mathcal{P}$ , compute
      $d_i = \frac{1}{|\mathcal{E}_{\text{sel}}|} \sum_{\mathbf{e}_{\text{sel}} \in \mathcal{E}_{\text{sel}}} \|\mathbf{e}_i - \mathbf{e}_{\text{sel}}\|_2$ 
7   Select triplet with maximum diversity:
      $(\alpha_{i^*}, a_{i^*}, \mathbf{e}_{i^*}) = \arg \max_{(\alpha_i, a_i, \mathbf{e}_i) \in \mathcal{P}} d_i$ 
8   Update ensemble:  $\mathcal{A}_{\text{best}} \leftarrow \mathcal{A}_{\text{best}} \cup \{\alpha_{i^*}\}; \quad \mathcal{E}_{\text{sel}} \leftarrow \mathcal{E}_{\text{sel}} \cup \{\mathbf{e}_{i^*}\}$ 
9    $\mathcal{P} \leftarrow \mathcal{P} \setminus \{(\alpha_{i^*}, a_{i^*}, \mathbf{e}_{i^*})\}$ 
10 end
11 return  $\mathcal{A}_{\text{best}}$ 

```

Fig. 3: Surrogate-Assisted Ensemble Construction.

In the first phase, we generate a large set of random candidate architectures $\mathcal{A}_{\text{cand}}$ via discrete sampling from the search space. We then use the dual-surrogate framework to predict the accuracy a and the latent diversity embedding $\mathbf{e}$ for each candidate. To ensure the base quality of the ensemble, we apply an

accuracy threshold λ , forming a pruned candidate pool $\mathcal{P}$ that contains only high-performing models.

The second phase employs a greedy forward-selection strategy to maximize architectural diversity. Starting with the most accurate model as an anchor, the algorithm iteratively selects subsequent architectures that exhibit the maximum average Euclidean distance from the currently selected set in the latent space. By leveraging the geometric organization of the latent space provided by $\mathbf{f}_{\text{div}}$, this selection process explicitly favors architectures with dissimilar prediction patterns. This approach effectively identifies a diverse set of high-performing models, balancing individual predictive power with collective ensemble robustness.

4 Computational Experiment

In this section we present the experimental results as well as the metrics used for comparison. In Section 4.1, we describe the dataset collection procedure. Section 4.2 details the training setup for the surrogate functions. Finally, in Section 4.3, we compare our proposed method against DeepEnsemble [6] and Random Search [19] for FashionMNIST, CIFAR10 and CIFAR100, respectively.

4.1 Construction of the Training Dataset for Surrogate Models

The models in our dataset were trained according to the Fig. 4.

Table 1: Training Hyperparameters and Performance per Dataset

Dataset	Num. Cells	Initial Width	Num. Epochs	Avg. Acc. (%)	Avg. Top-1 Agreement
FashionMNIST	3	16	125	89.6 ± 0.5	0.900 ± 0.004
CIFAR-10	8	16	200	75.8 ± 0.6	0.693 ± 0.006
CIFAR-100	8	16	200	37.6 ± 1.1	0.324 ± 0.008

For training the models, we adopt the hyperparameter configuration from [15], varying only the number of epochs, the number of cells, and the channel width.

Table 1 summarizes the training hyperparameters and results for all sampled models.

To enhance the efficiency of the candidate generation process, we implement an iterative surrogate-assisted refinement strategy. Initially, a preliminary set of $N_1 = 1000$ architectures is sampled to perform the baseline training of the accuracy surrogate $f_{\text{acc}}^{\theta_{\text{acc}}}$. Following this, we filter the search space by retaining only the top 10% of candidates based on their predicted accuracy.

This is followed by the generation of an additional $N_2 = 2000$ architectures within the high-performance regions identified by the surrogate.

Input: $\mathcal{A}_{DARTS}$: architectures available through DARTS;
 $\mathcal{X}$: dataset for train and validate architectures;
 N : number of architectures.
Output: Model dataset $\mathcal{D} = \{(\alpha_i, \mathbf{y}_i, \text{acc}_i)\}_{i=1}^N$.

- 1 Split dataset $\mathcal{X}$ into training and validation subsets: $\mathcal{X}_{\text{train}}, \mathcal{X}_{\text{val}}$ with a 20%/80% ratio
- 2 **for** $i \leftarrow 1$ **to** N **do**
- 3 Sample architecture $\alpha_i \sim \text{Uniform}(\mathcal{A}_{DARTS})$
- 4 Train model $f_{\alpha_i}(\mathcal{X}_{\text{train}}, \omega(\alpha_i))$
- 5 Evaluate f_{α_i} on $\mathcal{X}_{\text{val}}$
- 6 prediction vector $\mathbf{y}_i = f_{\alpha_i}(\mathcal{X}_{\text{val}}, \omega^*(\alpha_i))$
- 7 validation accuracy

$$\text{acc}_i = \frac{1}{|\mathcal{X}_{\text{val}}|} \sum_{(x,y) \in \mathcal{X}_{\text{val}}} \mathbb{I}(\arg \max_c f_{\alpha_i}(x; \omega^*(\alpha_i)) = y)$$
- 8 $\mathcal{D} \leftarrow \mathcal{D} \cup (\alpha_i, \mathbf{y}_i, \text{acc}_i)$
- 9 **end**
- 10 **return** $\mathcal{D}$

Fig. 4: Construction of the Model Evaluation Dataset

4.2 Training of the Surrogate Functions

In our experiment, each cell contains $n = 5$ nodes, with two outgoing edges per node assigned random operations from $\mathcal{O}$, yielding $m = 10$ edges per cell. The operation set $\mathcal{O}$ includes: separable convolutions (3×3 , 5×5); dilated separable convolutions (3×3 , 5×5); 3×3 max pooling; 3×3 average pooling; Skip connection

We employed a graph attention network (GAT) with four convolutional layers and two fully connected layers at the output. The training of the surrogate functions used a cosine annealing learning rate scheduler.

Full models are constructed by stacking normal and reduction cells in a 2:1 ratio, starting from a predefined number of initial channels. Each model combines one normal and one reduction cell sampled independently from $\mathcal{A}$.

4.3 Ensemble Performance Across Datasets

We evaluate the proposed Surrogate Ensemble approach on three benchmark datasets of increasing complexity: FashionMNIST, CIFAR-10, and CIFAR-100. For each dataset, results are reported over multiple independently trained ensembles and compared against DARTS-based Deep Ensembles and Random Search baselines. Specifically, for each baseline we train five independent ensembles: on FashionMNIST, each ensemble consists of three models, while for all other datasets, ensembles are composed of five models. The corresponding quantitative results are summarized in Table 2.

FashionMNIST. On FashionMNIST, all methods achieve strong performance, reflecting the relatively low structural complexity of the dataset. The Surrogate

Table 2: Ensemble performance on FashionMNIST, CIFAR-10, and CIFAR-100. Results are reported as mean $\pm$ standard deviation over independent ensembles.

Dataset	Metric	Surrogate Ens.	DeepEns	Random Search
FashionMNIST	Top-1 Acc. (%)	95.3 $\pm$ 0.1	95.02 $\pm$ 0.35	95.01 $\pm$ 0.19
	Avg. Model Acc. (%)	94.7 $\pm$ 0.1	94.55 $\pm$ 0.43	94.32 $\pm$ 0.24
	NLL	0.263 $\pm$ 0.003	0.266 $\pm$ 0.006	0.265 $\pm$ 0.003
	Oracle NLL	0.199 $\pm$ 0.010	0.206 $\pm$ 0.003	0.191 $\pm$ 0.010
	Brier Score	0.089 $\pm$ 0.001	0.091 $\pm$ 0.006	0.092 $\pm$ 0.002
	ECE	0.124 $\pm$ 0.002	0.121 $\pm$ 0.001	0.120 $\pm$ 0.004
	Ambiguity	0.0058 $\pm$ 0.0009	0.0047 $\pm$ 0.0009	0.0069 $\pm$ 0.0013
	Norm. Disagreement	0.065 $\pm$ 0.008	0.062 $\pm$ 0.008	0.077 $\pm$ 0.009
	Pred. Disagreement	0.130 $\pm$ 0.016	0.125 $\pm$ 0.016	0.154 $\pm$ 0.018
	FGSM AUC	0.058 $\pm$ 0.0025	0.056 $\pm$ 0.0036	0.056 $\pm$ 0.0012
	PGD AUC	0.023 $\pm$ 0.0018	0.024 $\pm$ 0.0029	0.022 $\pm$ 0.0010
CIFAR-10	Top-1 Acc. (%)	97.80 $\pm$ 0.08	97.64 $\pm$ 0.09	97.56 $\pm$ 0.11
	Avg. Model Acc. (%)	96.73 $\pm$ 0.10	96.71 $\pm$ 0.14	96.48 $\pm$ 0.09
	NLL	0.208 $\pm$ 0.003	0.210 $\pm$ 0.003	0.212 $\pm$ 0.002
	Oracle NLL	0.156 $\pm$ 0.002	0.154 $\pm$ 0.004	0.158 $\pm$ 0.003
	Brier Score	0.055 $\pm$ 0.001	0.056 $\pm$ 0.001	0.057 $\pm$ 0.001
	ECE	0.136 $\pm$ 0.002	0.135 $\pm$ 0.002	0.136 $\pm$ 0.001
	Ambiguity	0.011 $\pm$ 0.001	0.009 $\pm$ 0.001	0.011 $\pm$ 0.001
	Norm. Disagreement	0.044 $\pm$ 0.001	0.046 $\pm$ 0.003	0.046 $\pm$ 0.001
	Pred. Disagreement	0.087 $\pm$ 0.003	0.093 $\pm$ 0.007	0.092 $\pm$ 0.003
	FGSM AUC	0.0569 $\pm$ 0.0012	0.0555 $\pm$ 0.0021	0.0559 $\pm$ 0.0012
	PGD AUC	0.0273 $\pm$ 0.0015	0.0251 $\pm$ 0.0020	0.0256 $\pm$ 0.0018
CIFAR-100	Top-1 Acc. (%)	85.17 $\pm$ 0.16	85.10 $\pm$ 0.29	84.41 $\pm$ 0.09
	Avg. Model Acc. (%)	80.50 $\pm$ 0.12	80.97 $\pm$ 0.41	79.57 $\pm$ 0.19
	NLL	0.692 $\pm$ 0.007	0.682 $\pm$ 0.010	0.718 $\pm$ 0.006
	Oracle NLL	0.403 $\pm$ 0.004	0.406 $\pm$ 0.007	0.418 $\pm$ 0.006
	Brier Score	0.234 $\pm$ 0.002	0.232 $\pm$ 0.004	0.244 $\pm$ 0.002
	ECE	0.140 $\pm$ 0.004	0.133 $\pm$ 0.003	0.141 $\pm$ 0.002
	Ambiguity	0.047 $\pm$ 0.001	0.041 $\pm$ 0.002	0.048 $\pm$ 0.002
	Norm. Disagreement	0.206 $\pm$ 0.002	0.195 $\pm$ 0.005	0.215 $\pm$ 0.002
	Pred. Disagreement	0.411 $\pm$ 0.004	0.390 $\pm$ 0.010	0.430 $\pm$ 0.004
	FGSM AUC	0.037 $\pm$ 0.001	0.0368 $\pm$ 0.001	0.0358 $\pm$ 0.001
	PGD AUC	0.017 $\pm$ 0.001	0.0158 $\pm$ 0.001	0.0167 $\pm$ 0.0005

Ensemble attains the highest Top-1 accuracy (95.3%) and the best average individual model accuracy, indicating that the surrogate successfully identifies high-quality architectures. While Random Search exhibits the strongest diversity across all evaluated metrics (Ambiguity, Normalized Disagreement, and Predictive Disagreement), this diversity alone does not translate into superior ensemble accuracy. In fact, both DeepEns and Random Search yield statistically equivalent Top-1 performance despite differences in individual model quality.

CIFAR-10. On CIFAR-10, the benefits of the proposed approach become more pronounced. The Surrogate Ensemble achieves the highest Top-1 accuracy (97.80%) and the lowest negative log-likelihood, outperforming both baselines. Although individual model accuracies are comparable between Surrogate Ensemble and DeepEns, our method achieves substantially higher ensemble-level gains. This improvement is supported by increased diversity, as reflected by Ambiguity values comparable to Random Search, while maintaining stronger individual predictors. These results demonstrate that the dual-surrogate framework effectively balances accuracy and diversity when the search space provides sufficient architectural variability.

CIFAR-100. On the more challenging CIFAR-100 dataset, the Surrogate Ensemble continues to outperform both baselines at the ensemble level, achieving the highest Top-1 accuracy (85.17%) despite slightly lower average individual model accuracy than DeepEns. This discrepancy highlights a key advantage of the proposed method: rather than maximizing individual performance in isolation, the surrogate-guided selection prioritizes mutually complementary architectures. In contrast, Random Search attains the highest diversity metrics but suffers from consistently weaker individual models, leading to inferior ensemble performance. Overall, these results confirm that principled diversity, informed by latent architectural structure, is crucial for constructing effective ensembles in complex search spaces.

Overall, across all evaluated datasets, the proposed surrogate-guided ensemble construction consistently outperforms the considered baselines, demonstrating that explicitly accounting for architectural diversity is critical for achieving strong ensemble performance.

5 Conclusion

In this study, we showed that the proposed surrogate diversity function learns a latent embedding of model architectures in which inter-model distances consistently reflect high levels of predictive diversity across FashionMNIST, CIFAR-10, and CIFAR-100. Building on this observation, we proposed a surrogate-guided ensemble construction framework that jointly optimizes accuracy and diversity, demonstrating consistent improvements over standard baselines. Our results further indicate that the role of diversity becomes increasingly important as dataset complexity grows, highlighting the necessity of principled diversity modeling in challenging settings.

A key limitation of the proposed approach is the computational cost associated with training a large pool of base models to construct the surrogate dataset. Future work may mitigate this overhead through more efficient sampling strategies or weight-sharing mechanisms, as well as by replacing discrete similarity measures with continuous information-theoretic metrics to improve the smoothness and expressiveness of diversity estimation.

Bibliography

- [1] Lars Kai Hansen and Peter Salamon. Neural network ensembles. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(10):993–1001, 1990. <https://doi.org/10.1109/34.58871>.
- [2] Ye Ren, Le Zhang, and P. N. Suganthan. Ensemble classification and regression—recent developments, applications and future directions. *IEEE Computational Intelligence Magazine*, 11(1):41–53, February 2016. ISSN 1556-603X. <https://doi.org/10.1109/MCI.2015.2471235>.
- [3] Corinna Cortes, Xavier Gonzalvo, Vitaly Kuznetsov, Mehryar Mohri, and Scott Yang. AdaNet: Adaptive structural learning of artificial neural networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 874–883. PMLR, 06–11 Aug 2017. URL <https://proceedings.mlr.press/v70/cortes17a.html>.
- [4] Sheheryar Zaidi, Arber Zela, Thomas Elsken, Chris C. Holmes, Frank Hutter, and Yee Whye Teh. Neural ensemble search for uncertainty estimation and dataset shift. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems 34*, pages 7898–7911, 2021.
- [5] Zhichao Lu, Ran Cheng, Shihua Huang, Haoming Zhang, Changxiao Qiu, and Fan Yang. Surrogate-assisted multi-objective neural architecture search for real-time semantic segmentation. *CoRR*, abs/2208.06820, 2022. <https://doi.org/10.48550/ARXIV.2208.06820>.
- [6] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *NeurIPS*, volume 30, pages 6405–6416, 2017.
- [7] Minghao Chen, Jianlong Fu, and Haibin Ling. One-shot neural ensemble architecture search by diversity-guided search space shrinking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16530–16539, June 2021.
- [8] Wei Wen, Hanxiao Liu, Yiran Chen, Hai Li, Gabriel Bender, and Pieter-Jan Kindermans. Neural predictor for neural architecture search. In A. Vedaldi, H. Bischof, T. Brox, and J. Frahm, editors, *ECCV*, pages 660–676. Springer, 2020. https://doi.org/10.1007/978-3-030-58580-8_39.
- [9] Colin White, Willie Neiswanger, and Yash Savani. Bananas: Bayesian optimization with neural architectures for neural architecture search. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 10293–10301, 2021.
- [10] Yu Xue, Zhenman Zhang, and Ferrante Neri. Similarity surrogate-assisted evolutionary neural architecture search with dual encoding strategy. *Electronic Research Archive*, 32(2):1017–1043, 2024. ISSN 2688-1594. <https://doi.org/10.3934/era.2024050>.

- [11] Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. *CoRR*, abs/1710.10903, 2017. <https://doi.org/10.48550/arxiv.1710.10903>.
- [12] Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face recognition and clustering. In *CVPR*, pages 815–823, 2015. <https://doi.org/10.1109/CVPR.2015.7298682>.
- [13] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.
- [14] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009.
- [15] Hanxiao Liu, Karen Simonyan, and Yiming Yang. Darts: Differentiable architecture search. In *ICLR*. OpenReview.net, 2018.
- [16] Xuanyi Dong and Yi Yang. Nas-bench-201: Extending the scope of reproducible neural architecture search. *CoRR*, 2020. <https://doi.org/10.48550/arXiv.2001.00326>.
- [17] Chris Ying, Aaron Klein, Esteban Real, Eric Christiansen, Kevin Murphy, and Frank Hutter. Nas-bench-101: Towards reproducible neural architecture search. *CoRR*, abs/1902.09635, 2019. <https://doi.org/10.48550/arxiv.1902.09635>.
- [18] Tianle Cai, Shengjie Luo, Keyulu Xu, Di He, Tie-yan Liu, and Liwei Wang. Graphnorm: A principled approach to accelerating graph neural network training. In *International Conference on Machine Learning*, pages 1204–1215. PMLR, 2021.
- [19] Liam Li and Ameet Talwalkar. Random search and reproducibility for neural architecture search. In J. Peters and D. Sontag, editors, *UAI*, volume 124, pages 367–377. PMLR, 2020.